DSGE Model Estimation via VAR IRF Matching: The Short and Long of It

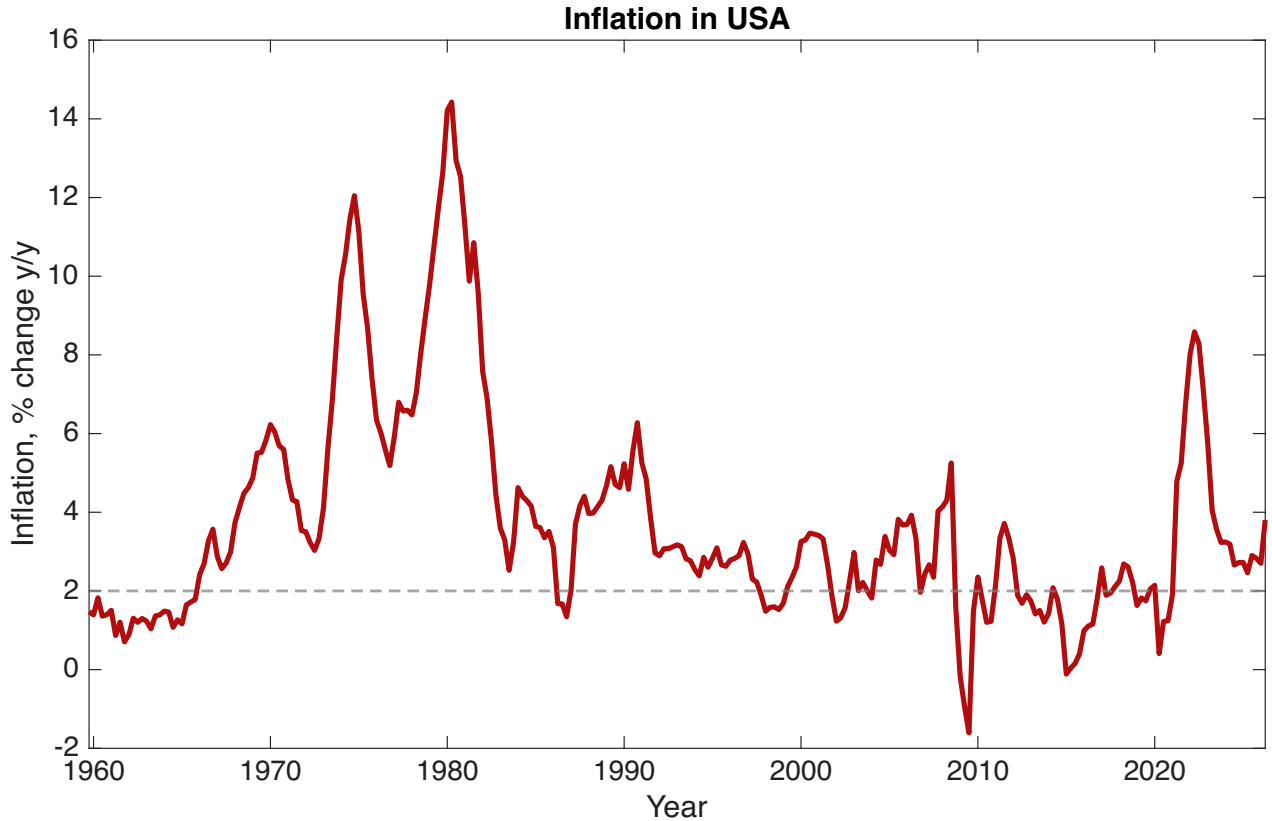
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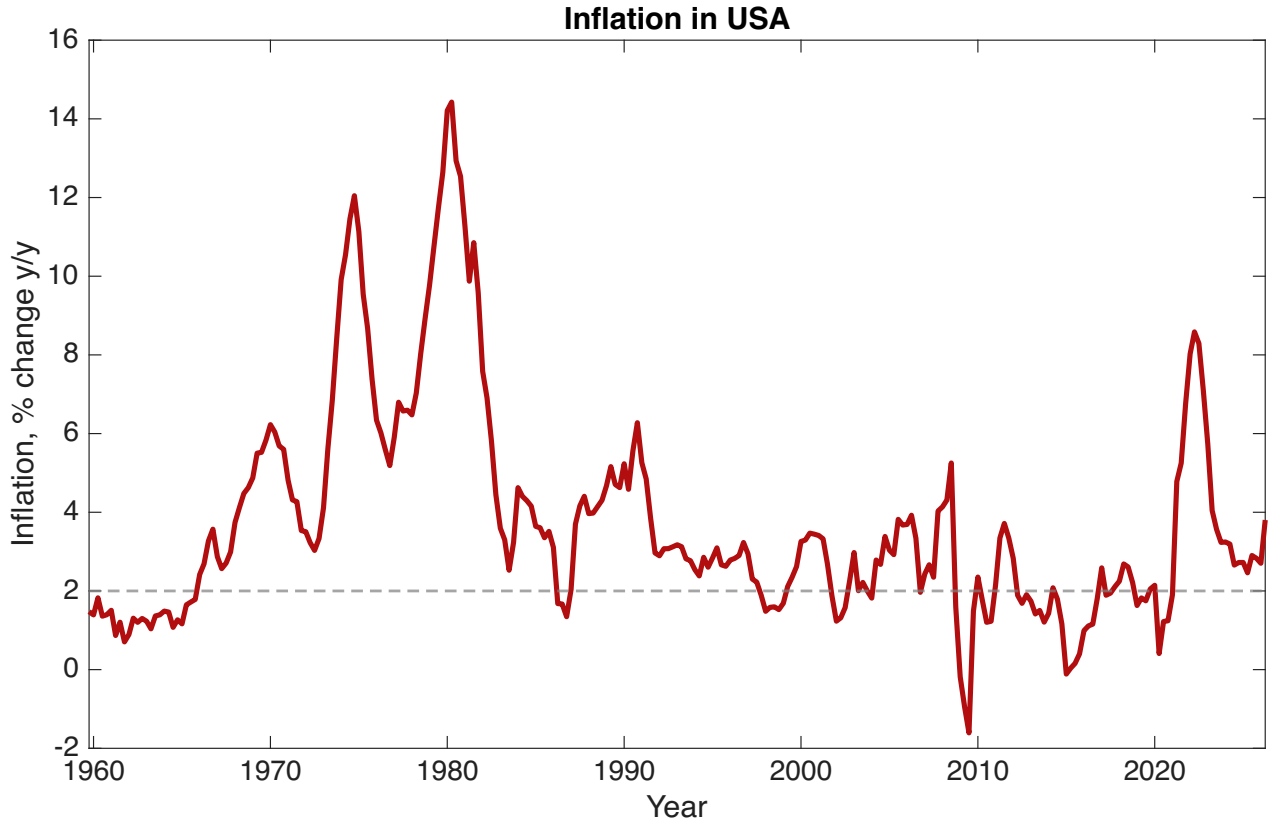
University of Padova
MSc Applied Economics

September 2026

Stylised Fact: Inflation Persistence

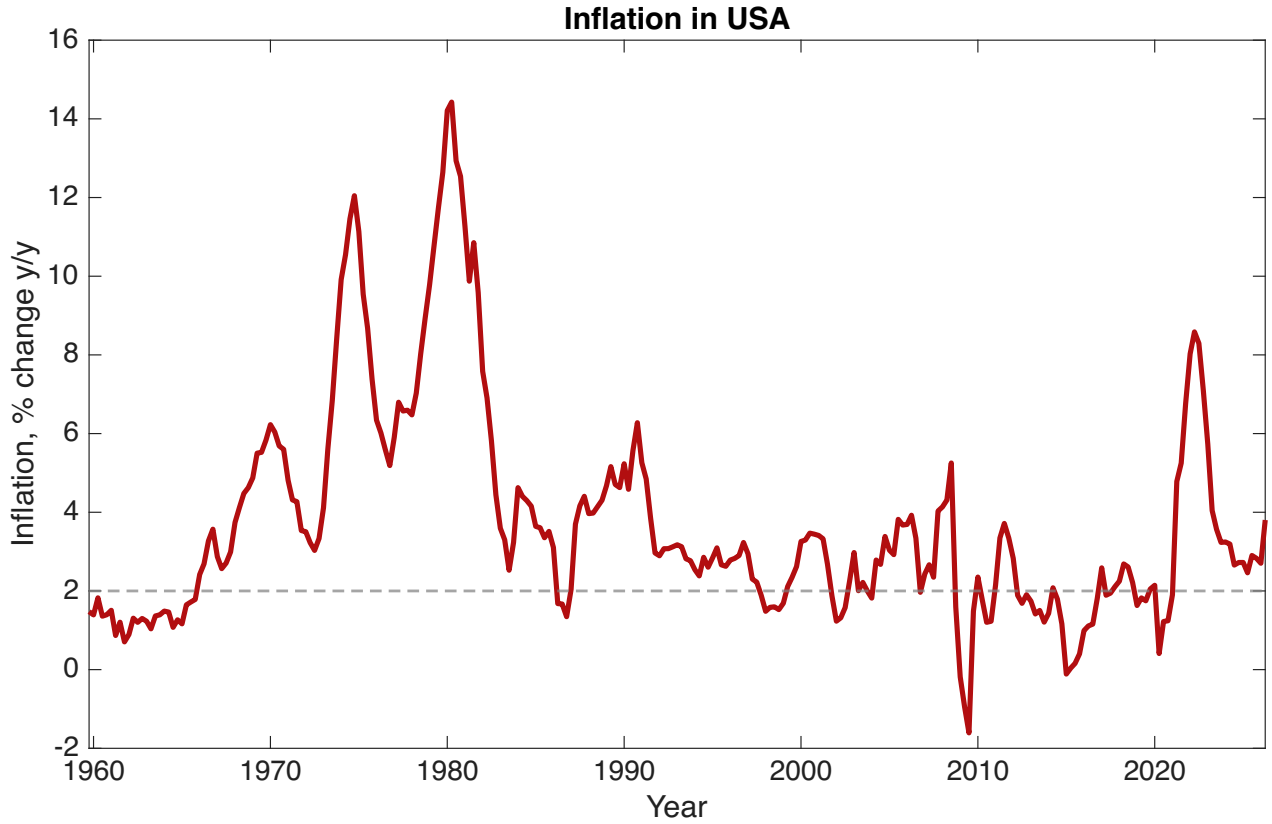


Stylised Fact: Inflation Persistence



- DSGE models are often used to explain inflation persistence

Stylised Fact: Inflation Persistence



- DSGE models are often used to explain inflation persistence
- But standard NK models reproduce it with **unrealistically high levels of price stickiness**, inconsistent with micro data evidence

CEE and ACEL's DSGE Motivation

- CEE JPE (2005) and ACEL RoED (2011) seminal works introduced the concept of **Firm Specific Capital** to model inflation persistence, mechanism supported by IRFs matching with VAR(4)

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- **Impulse Response Function Matching Estimation (IRFME)** is a common way to estimate DSGE models through VARs

$$\hat{\zeta} = \arg\min_{\zeta} (\hat{\Psi} - \Psi(\zeta))' V^{-1} (\hat{\Psi} - \Psi(\zeta)) \tag{1}$$

Motivation

- *If the* ***VAR is misspecified****, then, the* ***specification errors will be computed at each horizon****"*

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- **Truncation Bias is a type of misspecification** that arises when the VAR lag order is shorter respect its DGP.

- Misspecification can deliver distorted results misinforming the central banker

This Paper

Research Question:

- Can VAR **truncation bias** have significant implications for DSGE estimation?

What it's done:

- Revisit ACEL contribution, estimating Homogeneous vs Firm Specific Capital versions of their DSGE with VARs with different dynamic structures

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Findings:

- Increasing the VAR lag structure casts doubts on the importance of Specificity of Capital to explain inflation inertia
- Lag choice can have implications for optimal policy design, modelling and structural inference

Roadmap

- Related Literature
- ACEL model
- Truncation Bias
- Results

Related Literature

Long-Lag VARs:

- De Graeve and Westermark, (2025); Baumeister, (2025); Antolín-Díaz and Surico, (2025); Antolín-Díaz et al., (2026)

Truncation Bias:

- Ravenna, (2007); De Graeve and Westermark, (2025); Chari et al., (2008); Barnichon et al., (2026)

DSGE IRFs matching with VARs

- Rotemberg and Woodford (1997); Christiano et al., (2005); Boivin and Giannoni (2006); Altig et al., (2011); Rossi et al., (2012)

ACEL's DSGE Mechanism

ACEL build two models on Christiano et al. JPE, (2005):

- Homogeneous capital
- Firm-specific capital

- **Inflation is inertial** despite firms re-optimising prices on average every 1.5 quarters
- When firms re-optimise, **prices are adjusted by a small amount** because each firm's short run marginal cost is increasing in its own output
- The key assumption is that, in any given period, a **firm's capital stock is pre-determined and firm-specific**

ACEL's DSGE Mechanism (Cont'd)

- When log-linearized, the only difference in the characterizing equilibrium equations lies in the Phillips Curve

$$\Delta\hat{\pi}_t = \mathbb{E}[\beta\Delta\hat{\pi}_{t+1} + \gamma\hat{s}_t|\Omega_t] \tag{2}$$

$$\gamma = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}\chi \tag{3}$$

- **Models are observationally equivalent in terms of prices and quantities:** macro data identify γ, the difference between the two models lies in the implied ξ_p
- $\hat{s}_t$ is the economy-wide marginal cost, γ the Phillips Curve slope, ξ_p the Calvo price stickiness parameter
- χ reduces the sensitivity of inflation to marginal costs in the firm-specific capital model

ACEL's DSGE Mechanism (Cont'd)

Here χ is a non-linear function representing the strategic complementarity in price settings

- In the **Homogeneous** model, $\chi = 1$: γ depends only on β
- In the **Firm-specific** model, $\chi < 1$ and captures two forces that make a firm move its price *less* when it can reset:
  - It depends on Variable Capital Utilization σ_a
  - The costlier it is to vary capital utilization, the steeper the firm's own marginal cost curve
  - The more elastic the firm's demand, the larger the fall in output (and hence in marginal cost) after a price increase

- So χ governs **how much** a firm moves its price
- ξ_p **how often**, with implied duration $1/(1-\xi_p)$

Given a fixed value of γ, a smaller χ imply a smaller ξ_p

ACEL'S Monetary Policy Shock Propagation Mechanism

$\varepsilon_t^M \downarrow \Rightarrow i_t \downarrow \Rightarrow y_t^d \uparrow \Rightarrow y_t^s \uparrow \Rightarrow h_t^d \uparrow$
$\Rightarrow$ Calvo wages: $w_t \uparrow$, $\frac{w_t}{P_t} \uparrow$ little

Homogeneous

- $K_t = u_t \bar{K}_t \Rightarrow K_t^d \uparrow$ but $\hat{K}_t$ and $u_t \uparrow$ $r_t \uparrow$ given $\hat{\sigma}_a$
- $\Rightarrow MC_{i,t} = MC_t \Rightarrow$ MC independent from $y_{i,t}$ $\chi = 1$
- $MC_t \uparrow \Rightarrow P_t^{1-\xi_p} \uparrow \Rightarrow P_t \uparrow \Rightarrow \pi_t \uparrow$
- $\gamma = f(\bar{\xi}_p)$ given $\bar{\gamma}$ low, ξ_p need to be high enough

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Firm Specific

- $K_{i,t} = u_{i,t}\bar{K}_{i,t} \uparrow \Rightarrow K_t^d \uparrow$ but $\hat{K}$
- Given $\chi < 1 \Rightarrow MC_{i,t} \uparrow$ in $y_{i,t}$
- $MC_{i,t} \uparrow \Rightarrow P_t^{1-\xi_p} \uparrow \Rightarrow P_t \uparrow \Rightarrow \pi_t \uparrow$
- $\gamma = f(\bar{\xi}_p; \bar{\sigma}_a)$ given $\bar{\gamma}$

- If $\sigma_a \to 0 \Rightarrow \chi \to 1 \Rightarrow \gamma = f(\bar{\xi}_p; 0) \Rightarrow \xi_{p,\mathrm{FS}} \to \xi_{p,\mathrm{H}}$

ACEL (2005) Results

The estimated γ is 0.040, which implies:

- **Homogeneous** capital: $\xi_p = 0.82$, firms change prices every 5.6 quarters
- **Firm-specific** capital: firms change prices every 1.5 quarters

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- **Are these results robust to an enriched lag structure of the VAR?**

Illustration of the problem: Truncation Bias

$$Y_t = B_1 Y_{t-1} + \cdots + B_p Y_{t-p} + u_t, \qquad \mathbb{E}(u_t u_t') = \Sigma \tag{4}$$

$$B(L) = B_1 L + \cdots + B_p L^p \tag{5}$$

- The VAR with $p < p^*$ suffers from truncation bias.[1]
- Lag truncation biases the reduced form coefficients $B(L)$ and the reduced form variance covariance matrix Σ, which are the two main ingredients for IRF estimation.

$$\text{IRF} = f\big(B(L), \Sigma\big) \tag{6}$$

[1] Where p^* denotes the DGP true lag length.

Long-Lag VARs

De Graeve and Westermark, (2025) demonstrate that increasing the lag structure improves VAR's IRFs estimates obtaining:

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Lower Bias

- By reducing the truncation bias misspecification at each horizon

Lower Variance

- Misspecification reduction lowers the variance of the residuals
- This variance reduction counteracts the imprecision caused by increased parametrisation

Long-Lag VARs Benefits

The variance-reducing effect of lower misspecification often dominates the penalty of increased parameterisation.

Compared to short-lag VARs, Long-Lag VARs have:

- Reduced truncation bias
- More precise inference
- Lower MSE
- Better coverage

Residuals' Variance Reduction

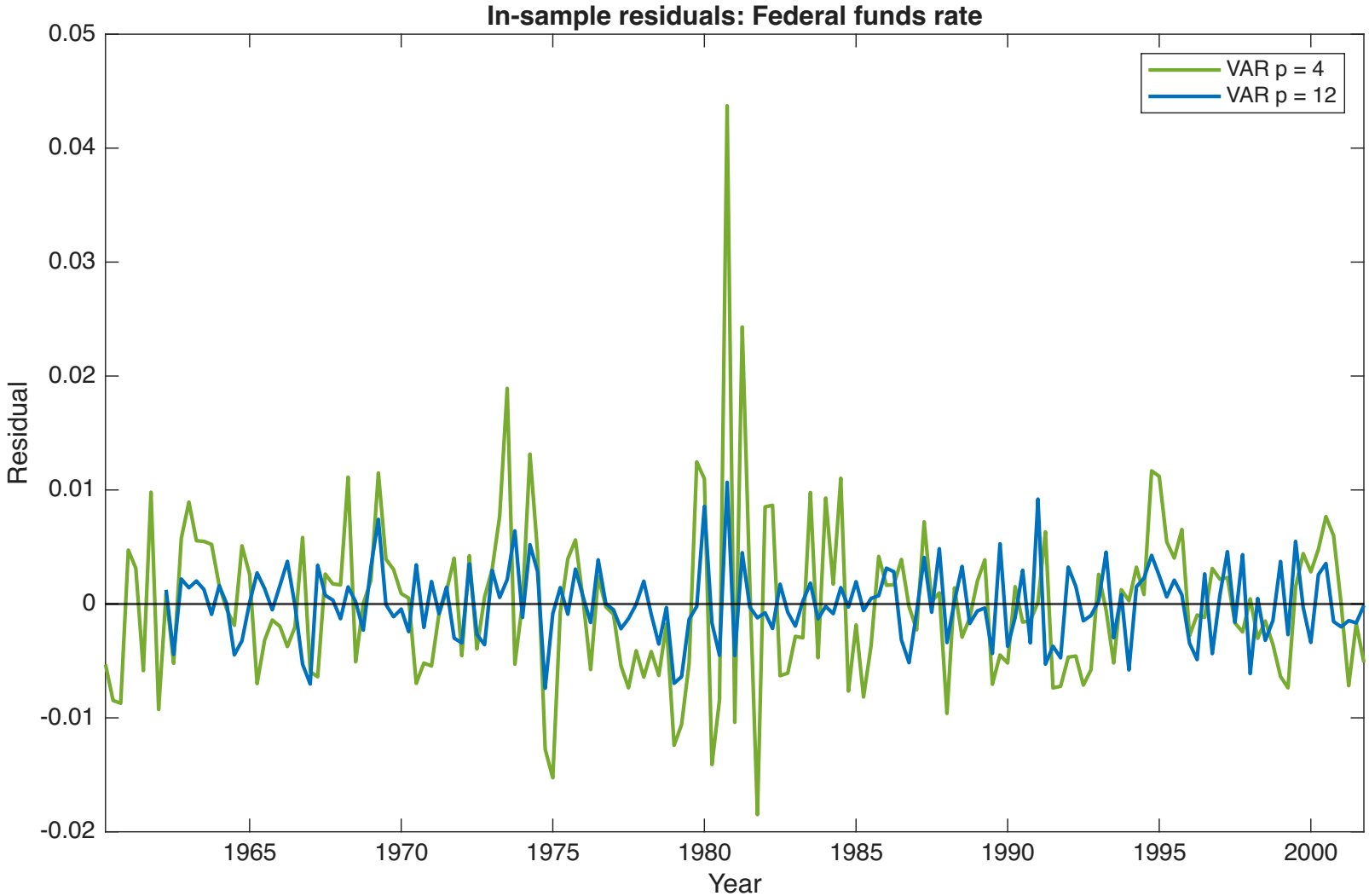
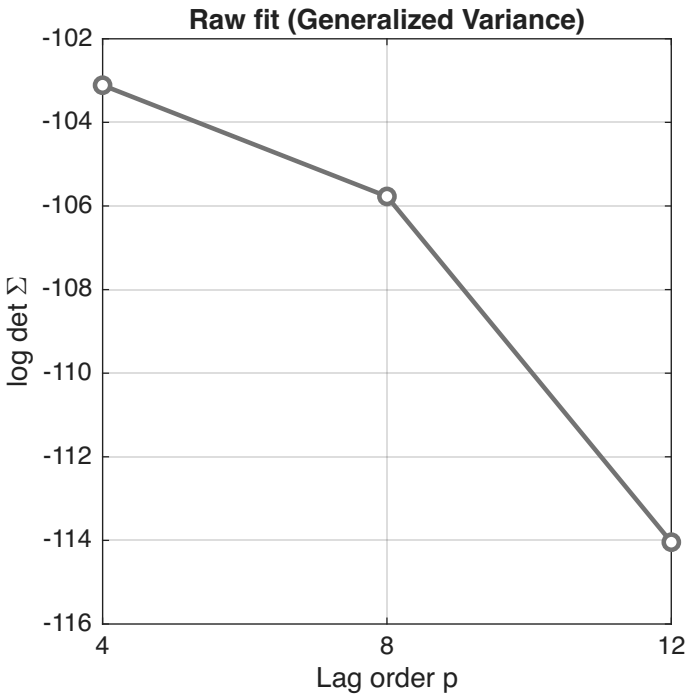
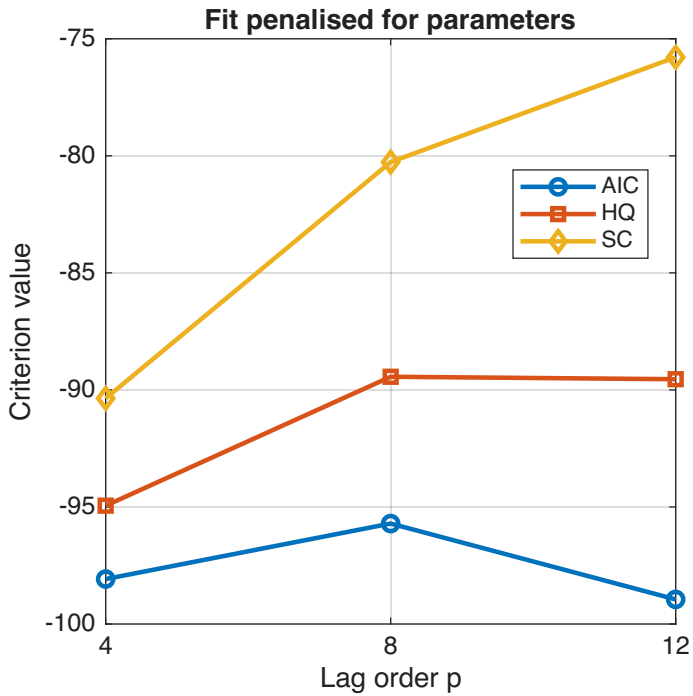


Figure 1: FFR Residuals Comparison Across lag 4 Vs 12

Generalised Variance and Information Criteria





Lag	$\log\det\hat{\Sigma}$	**AIC**	**HQ**	**SC**
$p = 4$	-103.1111	-98.0796	**-94.9444**	**-90.3591**
$p = 8$	-105.7716	-95.7087	-89.4383	-80.2677
$p = 12$	-114.0426	**-98.9483**	-89.5426	-75.7867

Replication Exercise

Framework:

- **Altig et al., (2011)**'s DSGE[2]

VAR:

- Short and Long-Run restrictions, medium scale 10 variables
- Monetary Policy, Neutral and Embodied Technology shocks
- Sample period 1959:Q2-2001:Q4

Exercise:

- I re-estimate the VAR increasing lag order
- Re-estimate the DSGE by IRFs matching
- **Analyse implications of possible variations in the structural parameters and IRFs across different lag choice**

[2]Where the replication package provided by the authors refers to the 2005 NBER working paper

VAR IRFs; Monetary Policy Shock

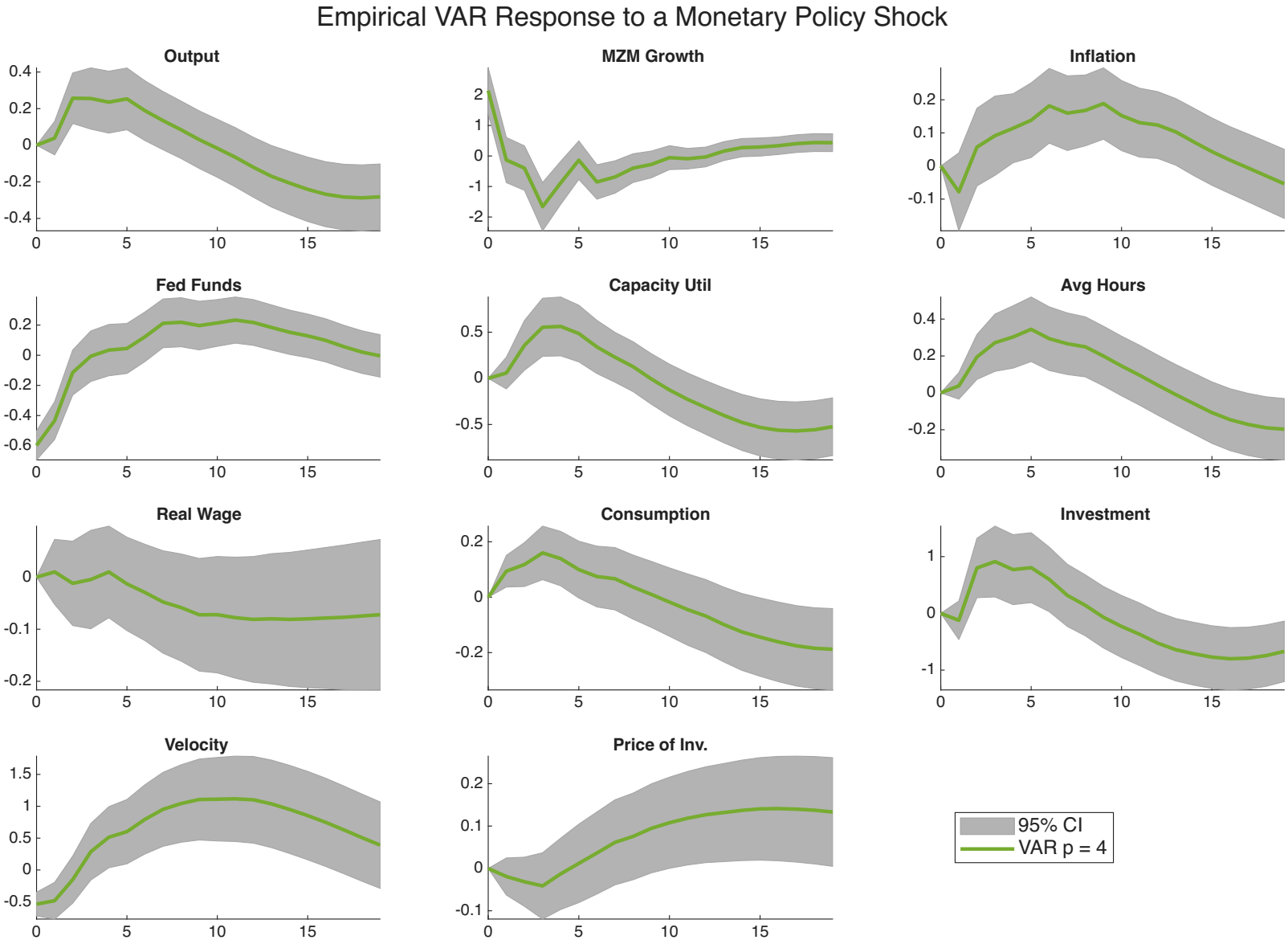


Figure 2: VAR IRFs Comparison: Monetary Policy Shock: 4 lags

VAR IRFs; Monetary Policy Shock

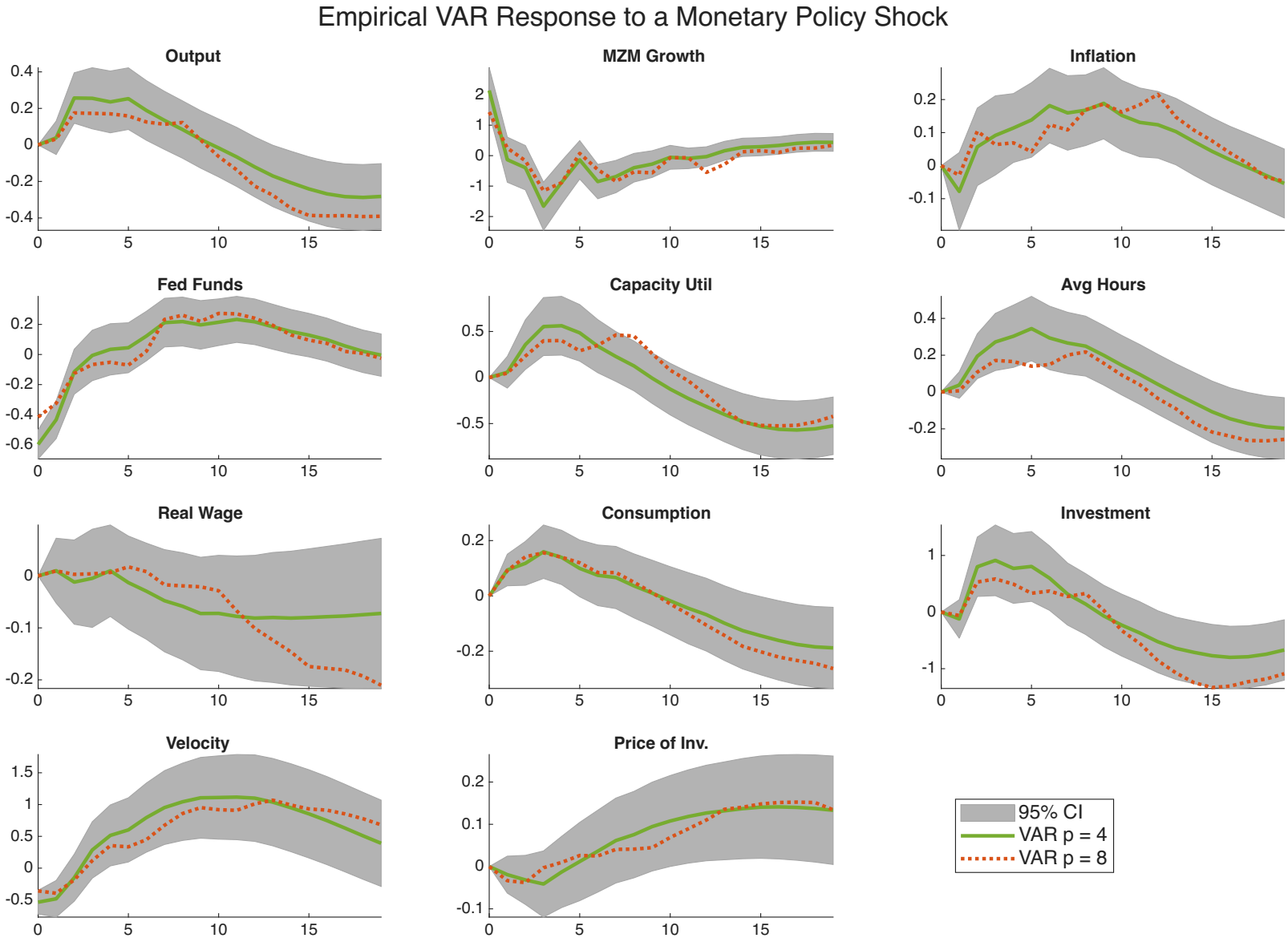


Figure 2: VAR IRFs Comparison: Monetary Policy Shock: 8 lags

VAR IRFs; Monetary Policy Shock

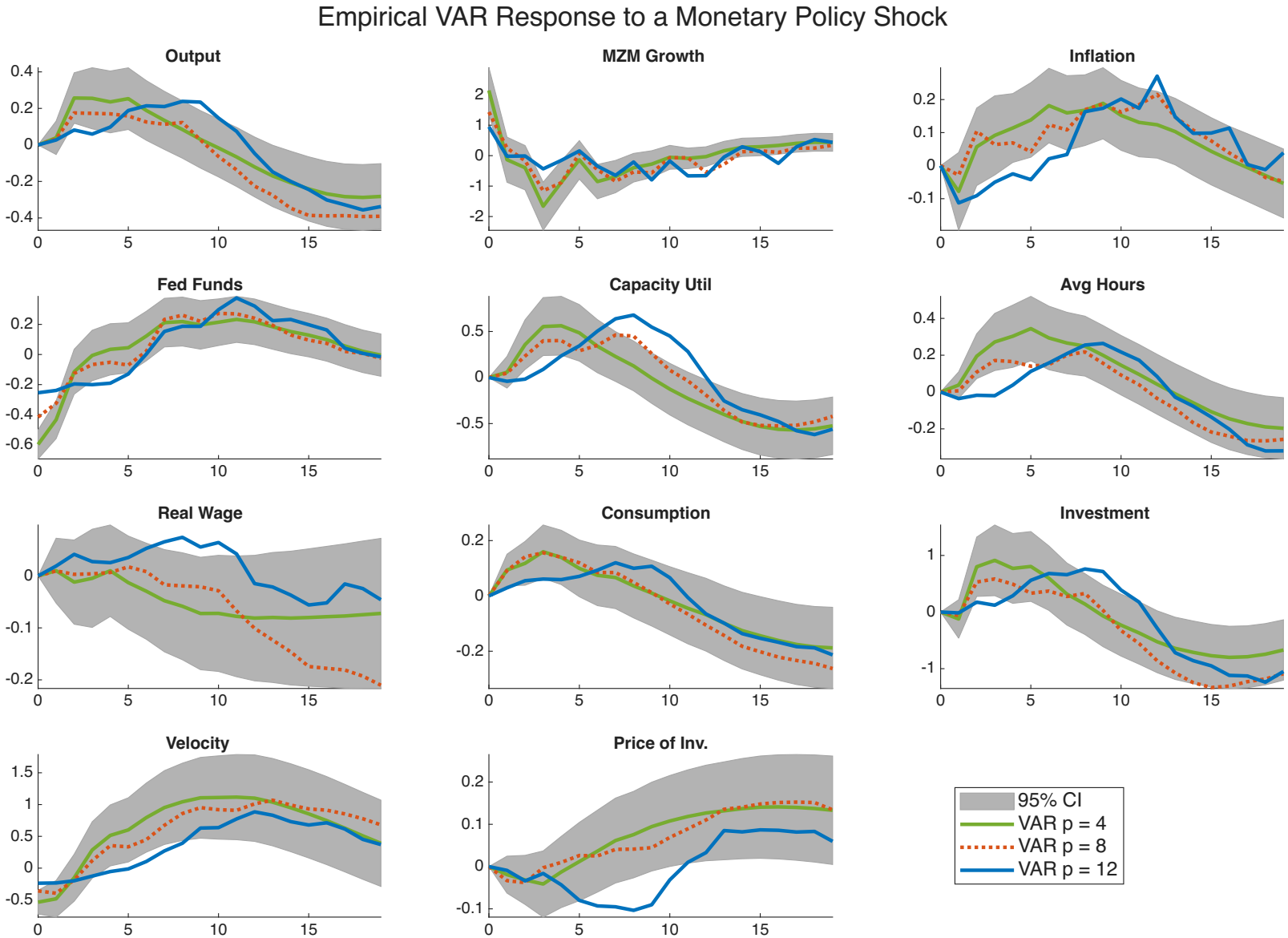


Figure 2: VAR IRFs Comparison: Monetary Policy Shock: 12 lags

VAR IRFs; Monetary Policy Shock

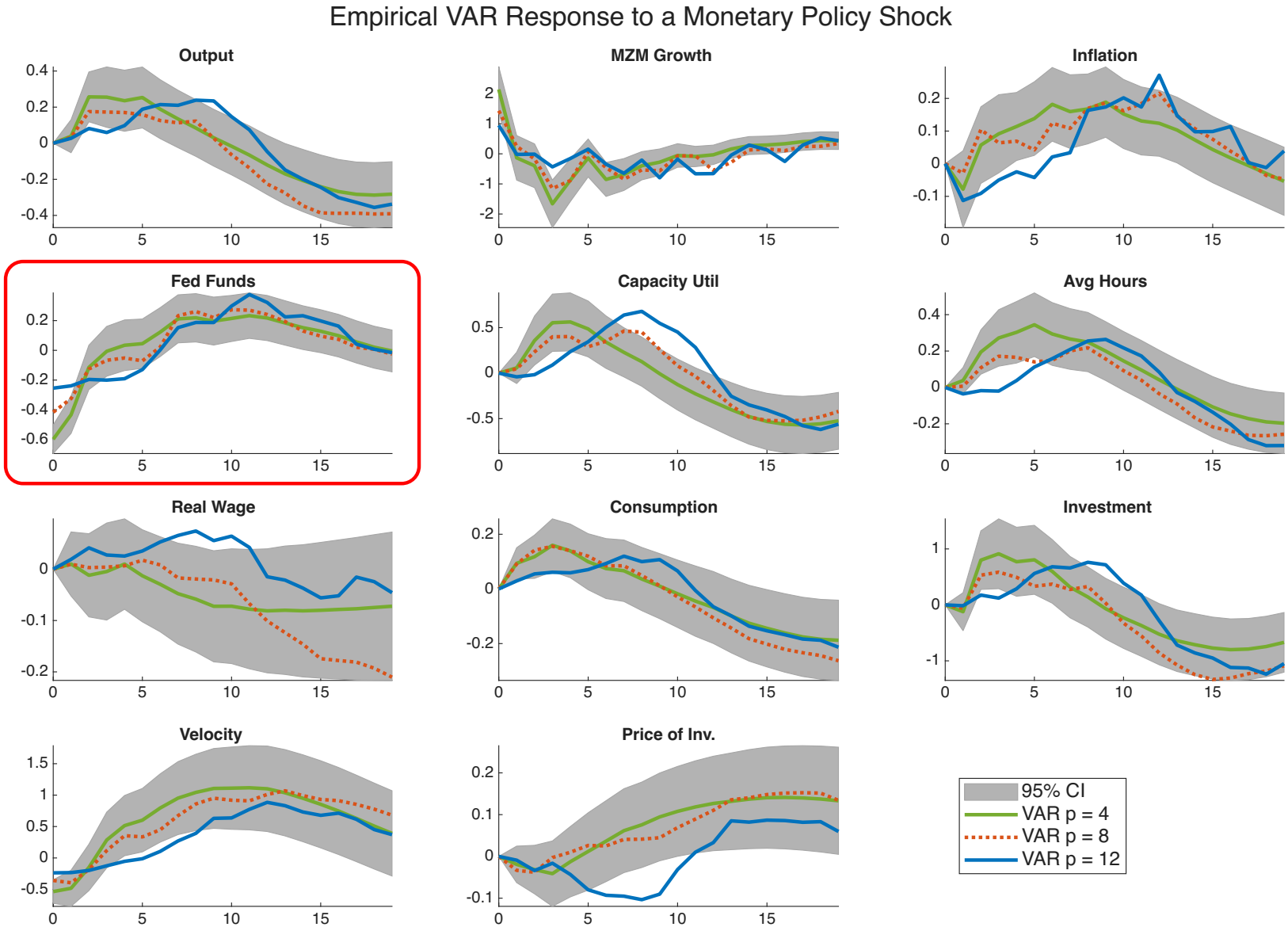


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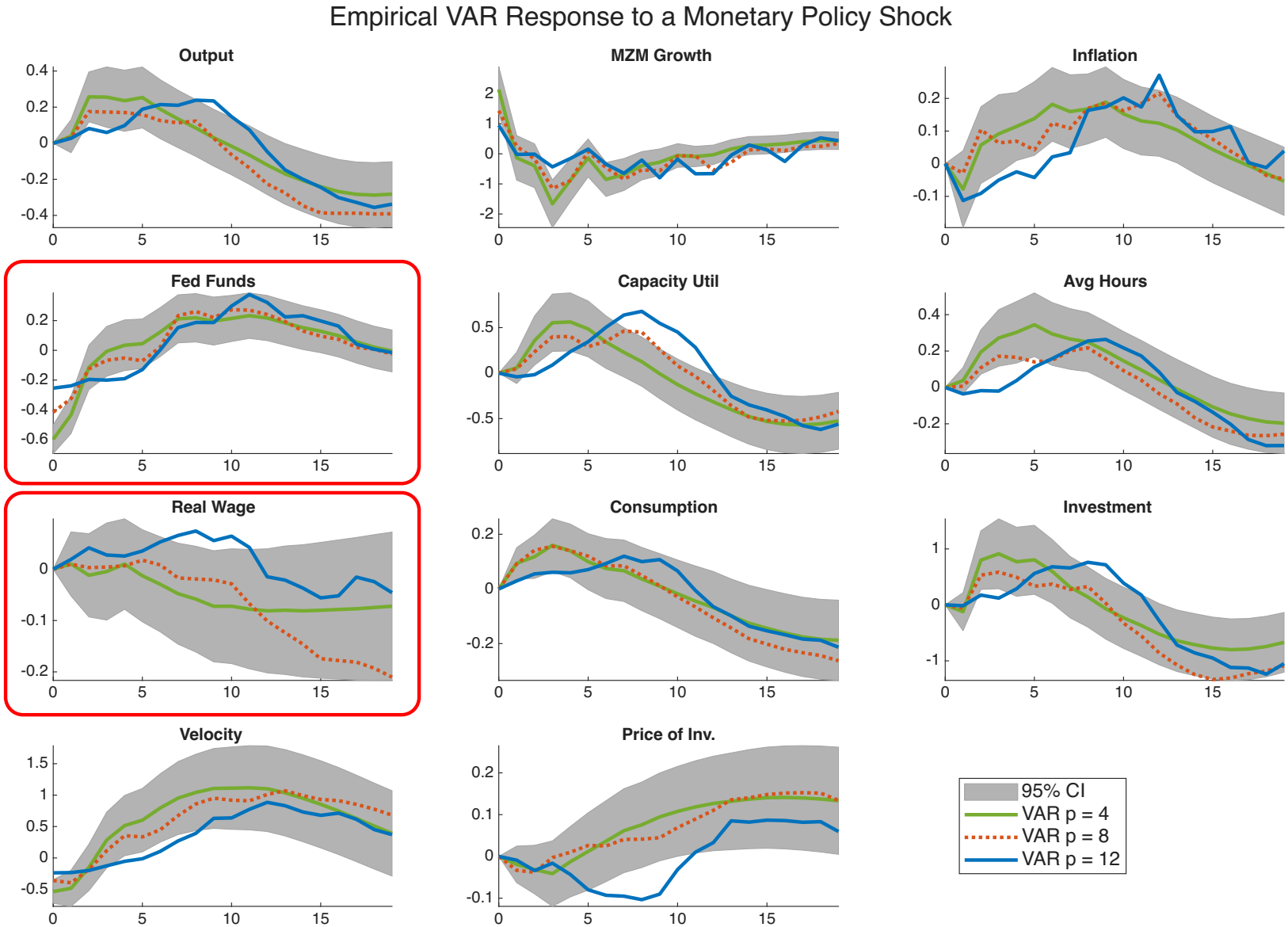


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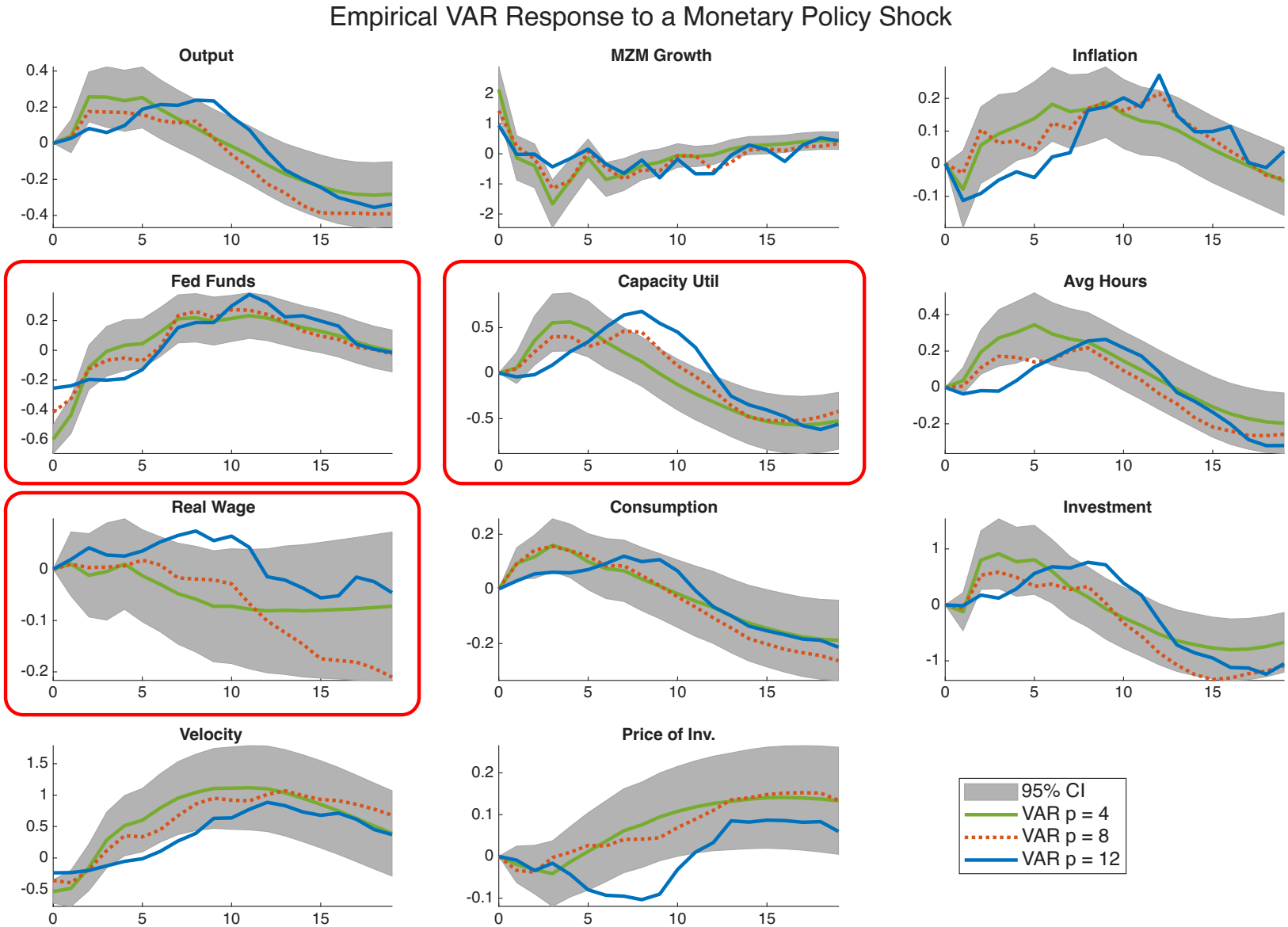


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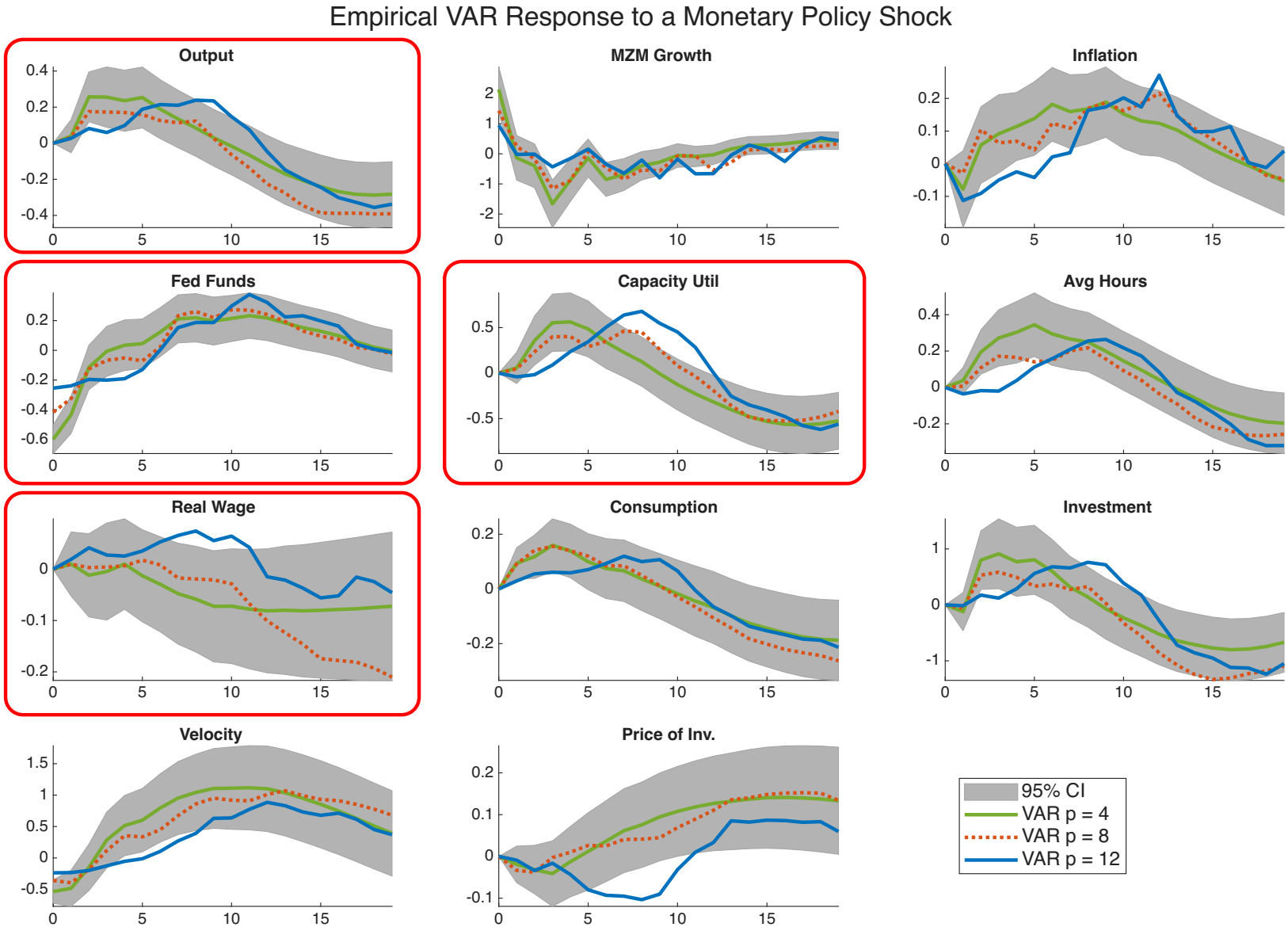


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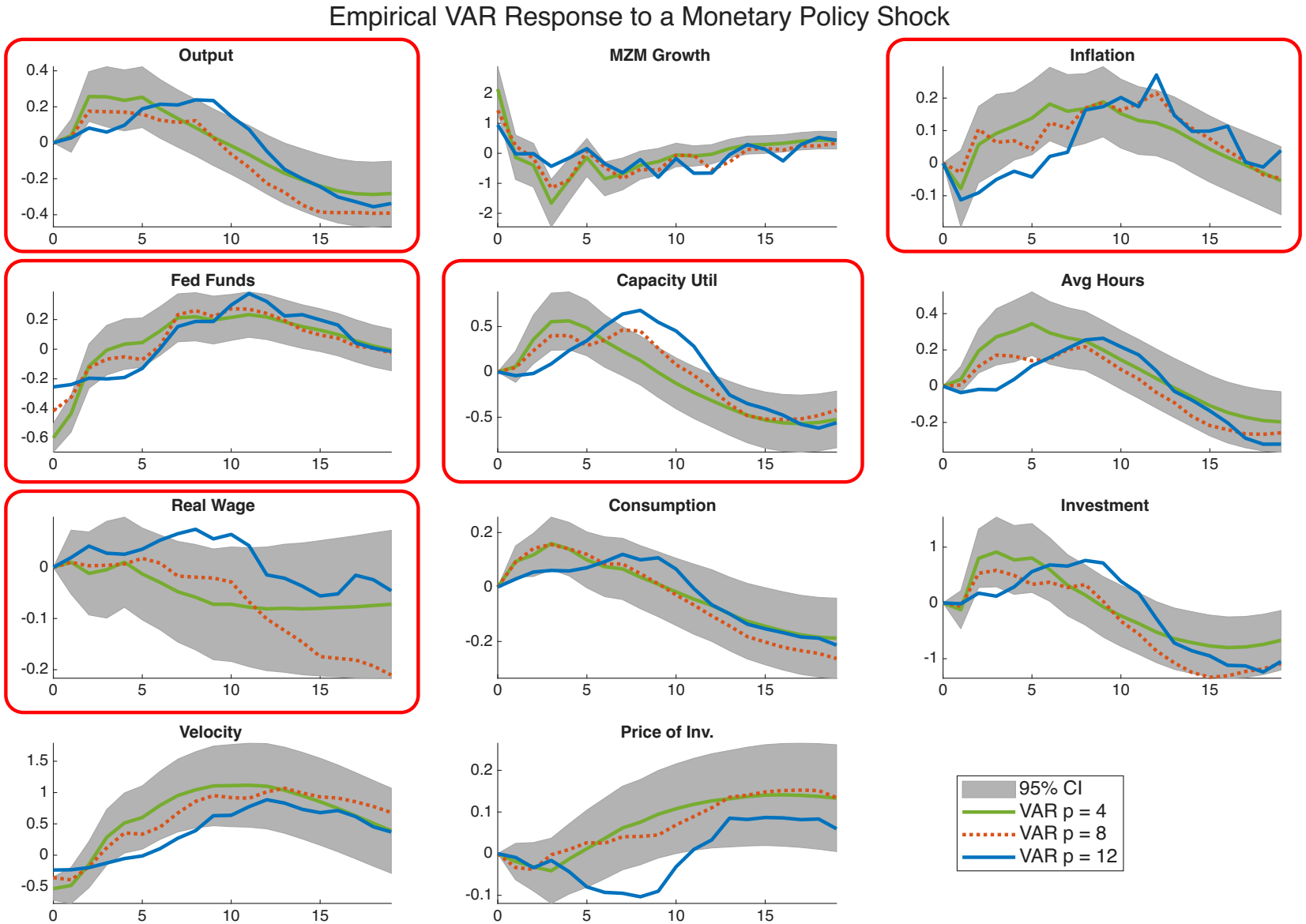


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DSGE Structural Parameters

MonteCalvo

Parameter	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Structural Estimated Frictions			
Price markup (λ_f)	1.0100	1.0100	1.0100
	(-)	(-)	(-)
Wage stickiness (ξ_w)	0.7221***	0.5617***	0.4838***
	(0.1506)	(0.1142)	(0.0757)
Phillips curve slope (γ)	0.0402	0.0402	0.0129***
	(0.0246)	(0.0598)	(0.0042)
Capacity util. cost (σ_a)	1.9938	0.1175	0.0042
	(2.6066)	(0.6299)	(0.0230)
Habit formation (b)	0.7066***	0.6935***	0.8596***
	(0.0463)	(0.0570)	(0.0203)
Inv. adj. cost (S'')	3.2807*	3.0372*	6.8431***
	(1.8874)	(1.5506)	(2.3306)
Money demand elast. (ϵ)	0.8077***	0.7531***	0.8346***
	(0.2111)	(0.1473)	(0.1944)
Implied Price Stickiness			
Homogeneous capital ($\xi_{p,hom}$)	0.8213***	0.8213***	0.8958***
	(0.0480)	(0.0574)	(0.0139)
Firm-specific capital ($\xi_{p,fs}$)	0.3372**	0.6448***	0.8884***
	(0.1505)	(0.1285)	(0.0220)

Note: Standard errors in parentheses. The Calvo parameters are recovered numerically from the estimated structural parameters; their standard errors come from a 1,000,000-draw Monte Carlo of the joint distribution of the estimated structural parameters.

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Implied Duration

Model Specification	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous Capital Model			
Calvo parameter ($\xi_{p,hom}$)	0.82	0.82	0.90
Implied duration (quarters)	**5.6**	**5.6**	**9.6**
Firm-Specific Capital Model			
Calvo parameter ($\xi_{p,fs}$)	0.34	0.64	0.89
Implied duration (quarters)	**1.5**	**2.8**	**9.0**

Note: Implied duration is calculated as $D = 1/(1 - \xi_p)$.

- Firm Specific model's Calvo Price parameter converges to the Homogeneous one
- Back to the micro-macro implied price duration mismatch

χ Across Lags

Conclusions

Following the advices from the Long-Lag VARs literature. Replicating ACEL et. al., (2005), enriching the VAR lag structure I find that:

- Capacity Utilization Cost, crucial variable to justify the firm specific model, dramatically shrinks to 0
- Firm specificity of capital seems not the key concept to explain inflation inertia
- ACEL's evidence on firm specific capital is not robust to long-lag VARs

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De Graeve & Westermark (2025) and Baumeister (2025)

- Long-Lag VARs might have structural implications, it's up to the researcher to demonstrate that their findings are robust to an enriched lag structure

Thanks a lot for your attention!

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Implied Real Rigidity (χ) Across Lags

Model	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
χ_{hom} $(\equiv 1)$	1.0000 (-)	1.0000 (-)	1.0000 (-)
χ_{fs}	0.0307 (0.0596)	0.2030 (0.1667)	0.8671*** (0.1698)

Note: Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. $\chi = \gamma\xi_p/[(1-\xi_p)(1-\beta\xi_p)]$, recovered by inverting the Phillips curve slope equation; $\chi_{hom} \equiv 1$ by construction. Standard errors for χ_{fs} come from the same 1,000,000-draw joint Monte Carlo used for the Calvo parameters.

Back to Implied Duration

VAR IRFs All CI; Monetary Policy Shock

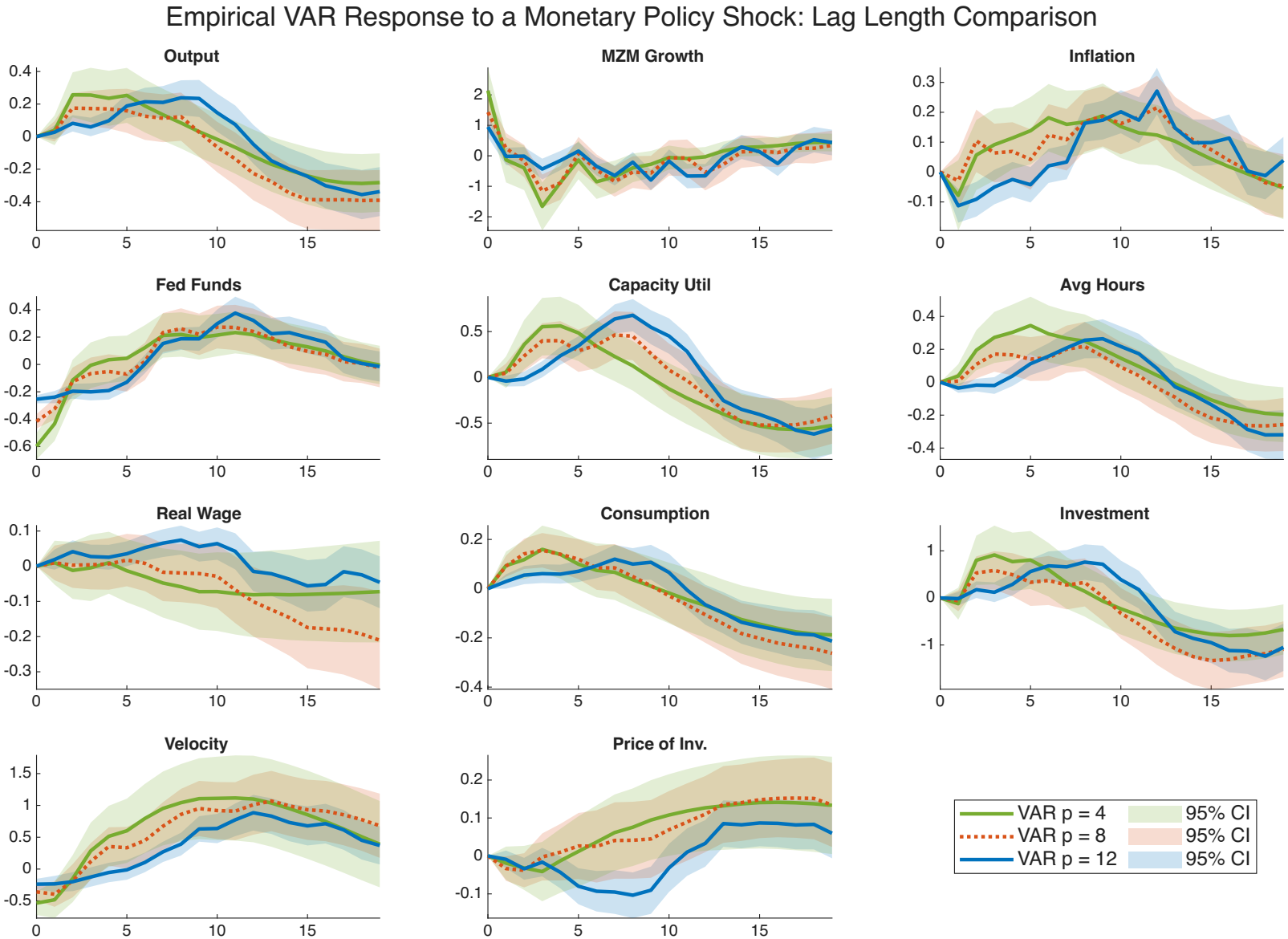


Figure 3: IRFs VARs all CI Comparison: Monetary Policy Shock

Bootstrap Confidence Intervals Shrinks!

- With respect to De Graeve and Westermark (2025) examples, this exercise has a medium scale VAR; 10 variables + constant
- Estimating the VAR at $p = 12$ we approach the feasible limit: only $T_u - k = 38$ degrees of freedom per equation, and a total number of coefficients $n + n^2 p = 1210$
- Bootstrap Confidence Intervals actually shrink!
- Confirming De Graeve and Westermark (2025) p.2 JME point on variance reduction even in a limit case "Misspecification reduction not only leads to Bias-Variance trade off, but it reduces variance"

DSGE IRFs; Monetary Policy Shock

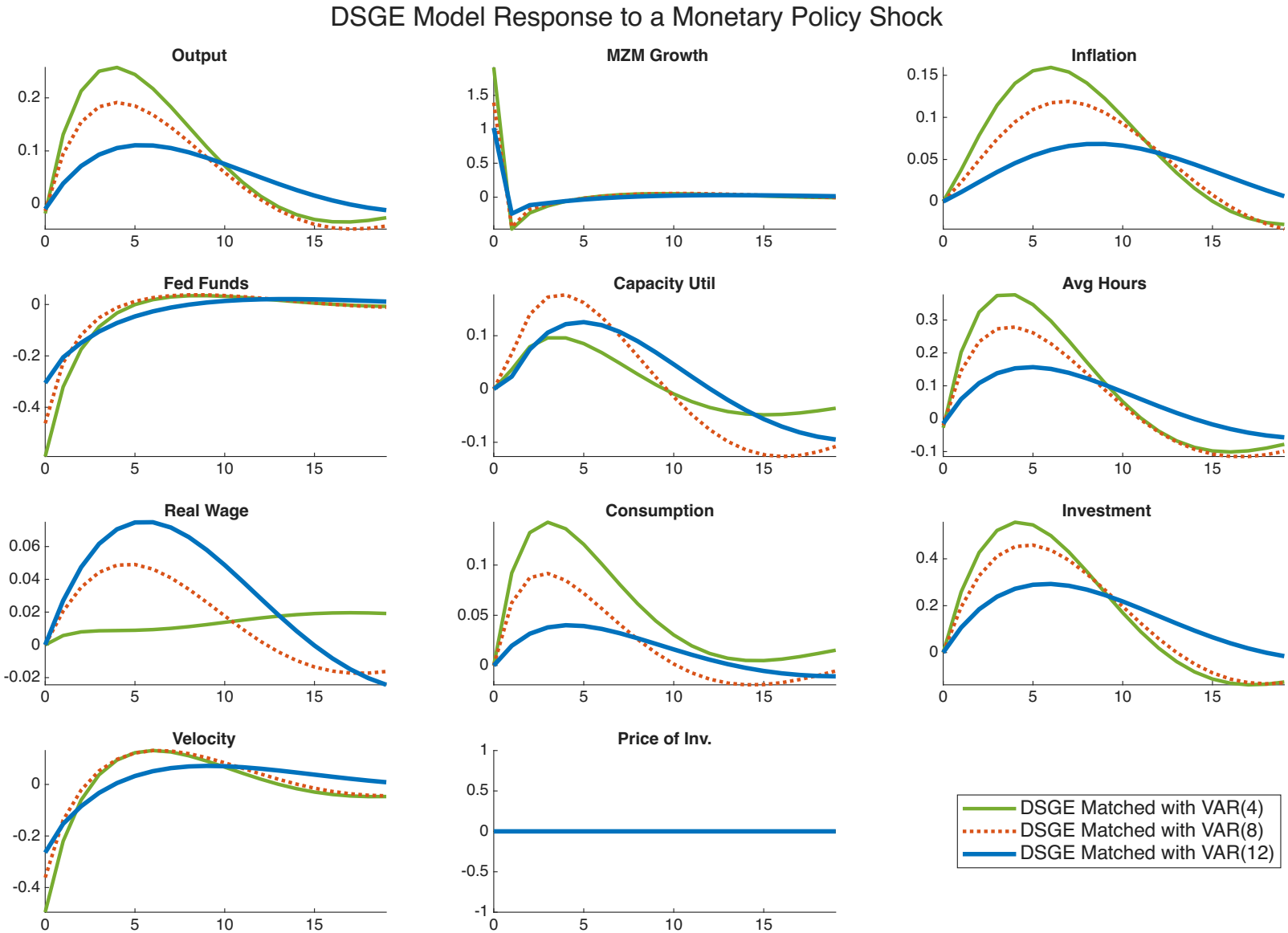


Figure 4: DSGE IRFs Long Lag Progressively Plots 4 8 12

VAR IRFs; Neutral Technology Shock

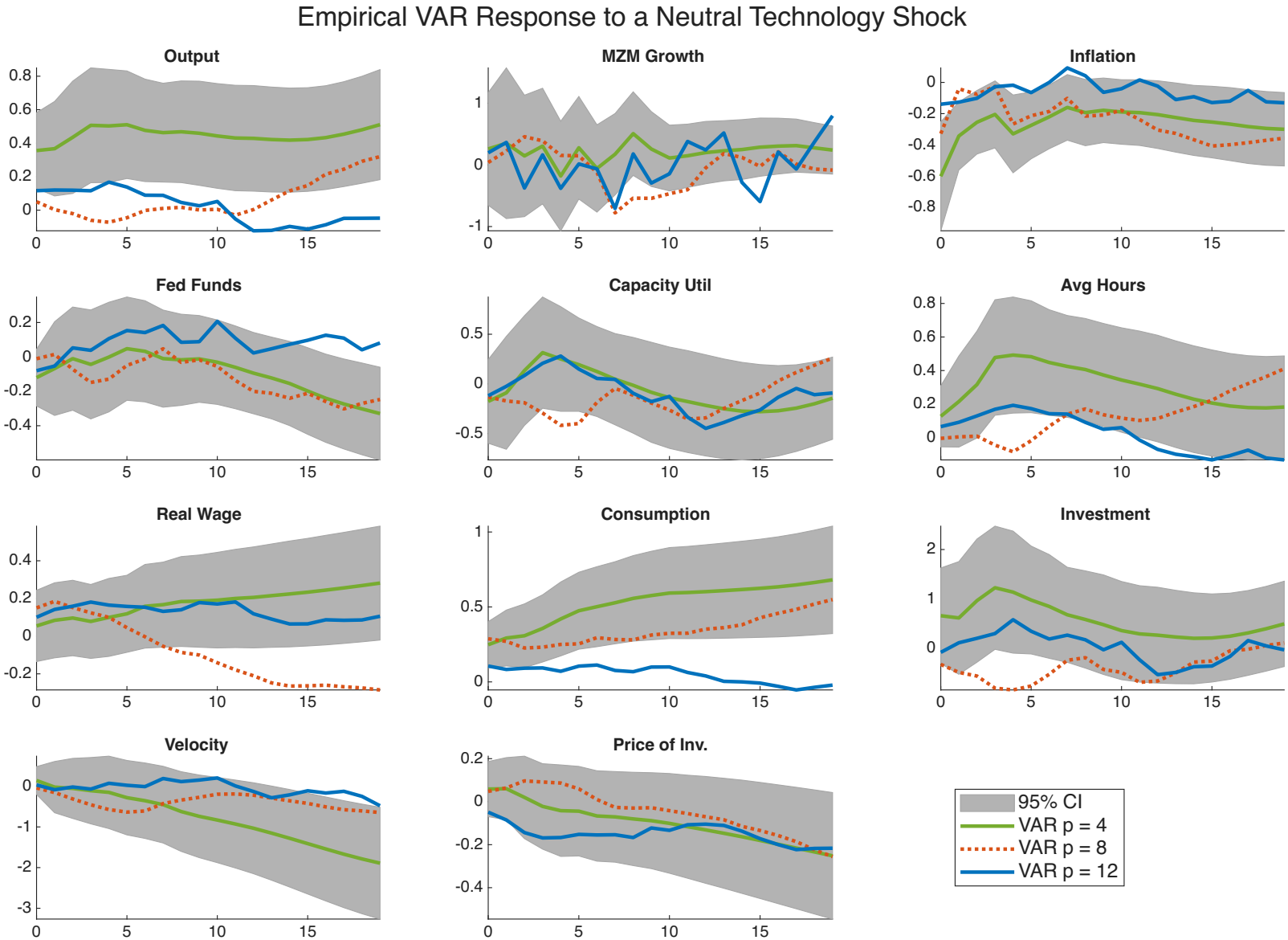


Figure 5: IRFs VARs all CI Comparison: Neutral Technology Shock

VAR IRFs; Embodied Technology Shock

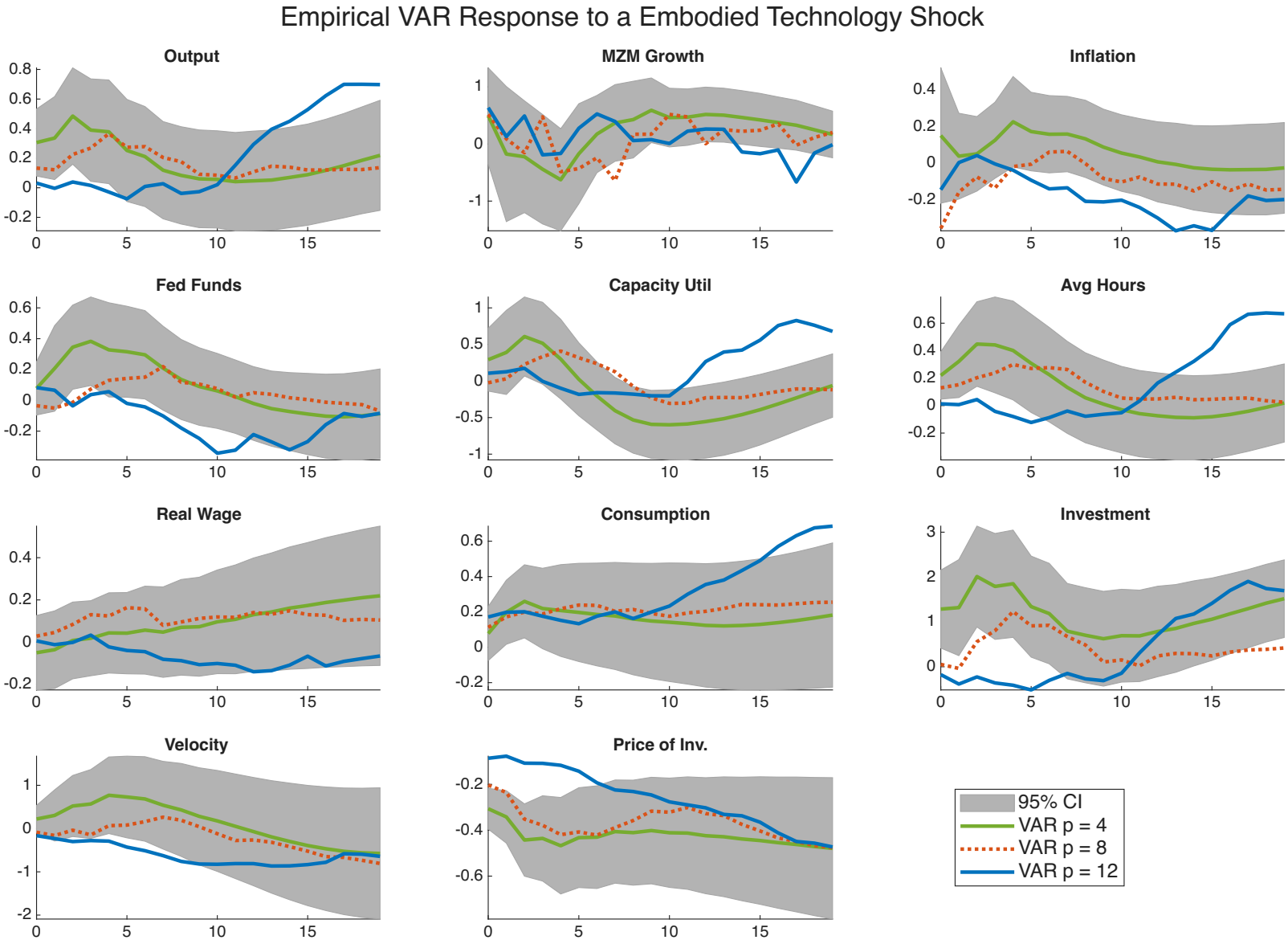


Figure 6: IRFs VARs all CI Comparison: Embodied Technology Shock

MonteCalvo

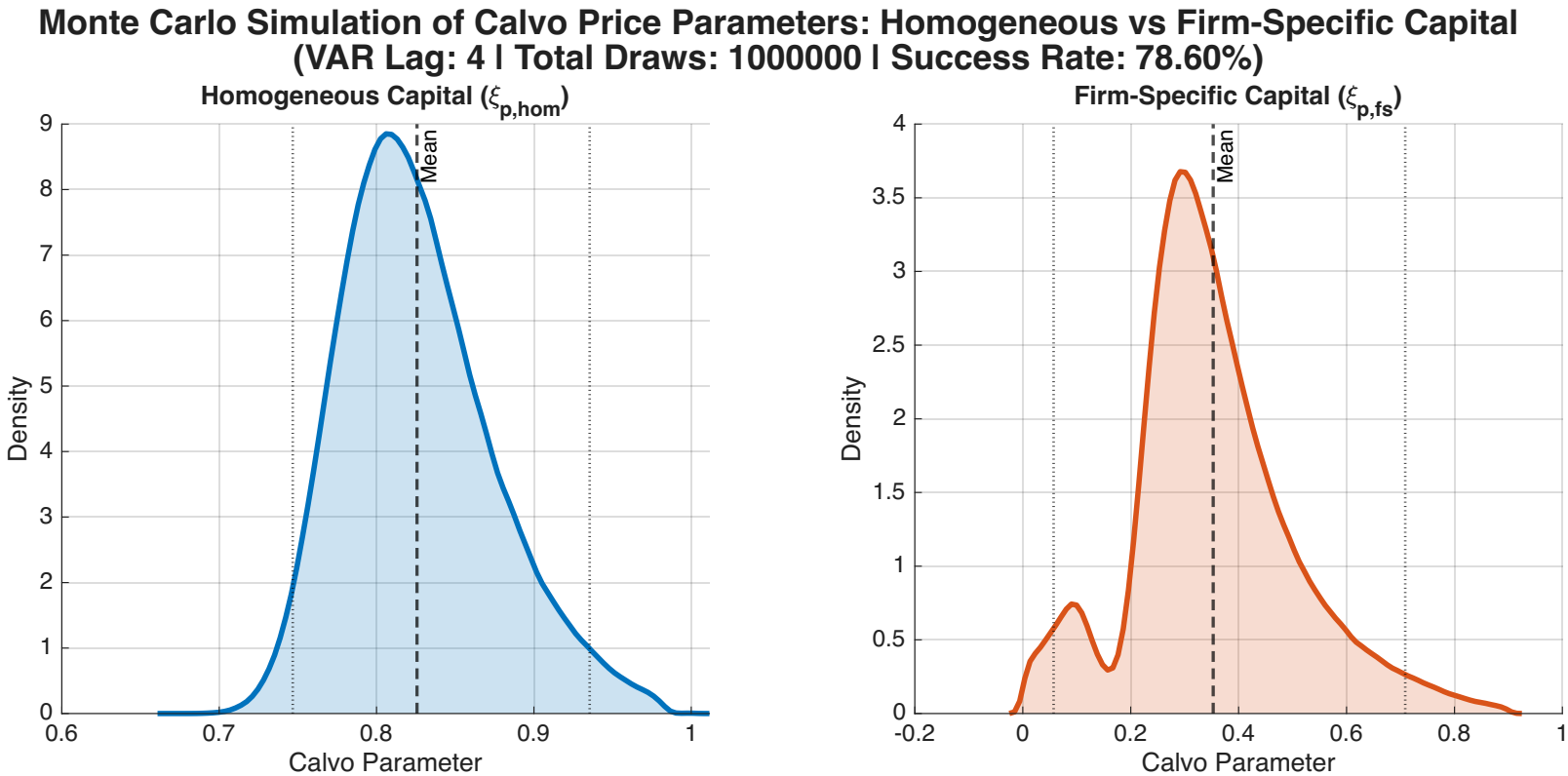


Figure 7: Calvo Price Parameter PDF from MonteCarlo Simulation; VAR(4)

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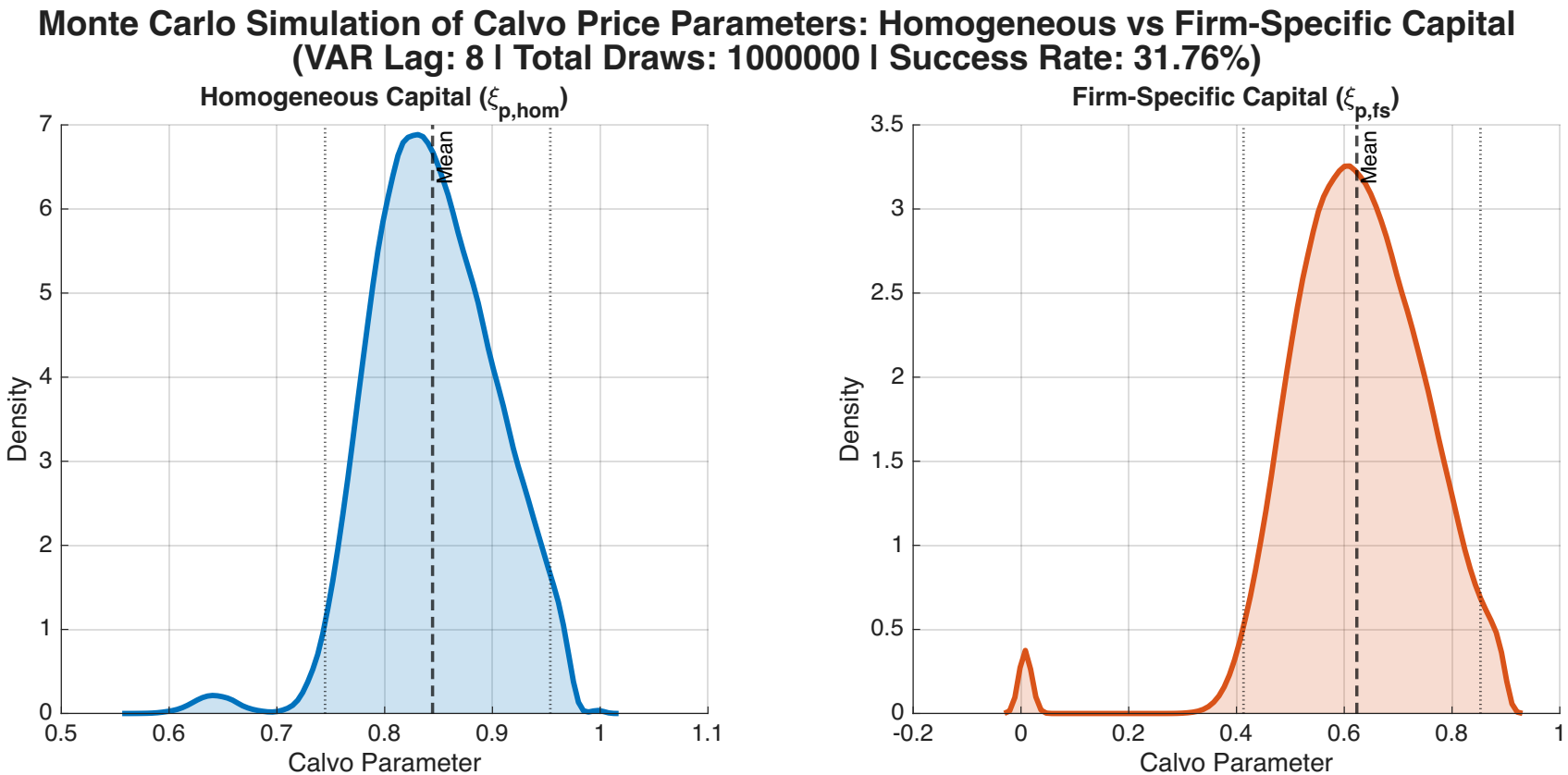


Figure 8: Calvo Price Parameter PDF from MonteCarlo Simulation; VAR(8)

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MonteCalvo

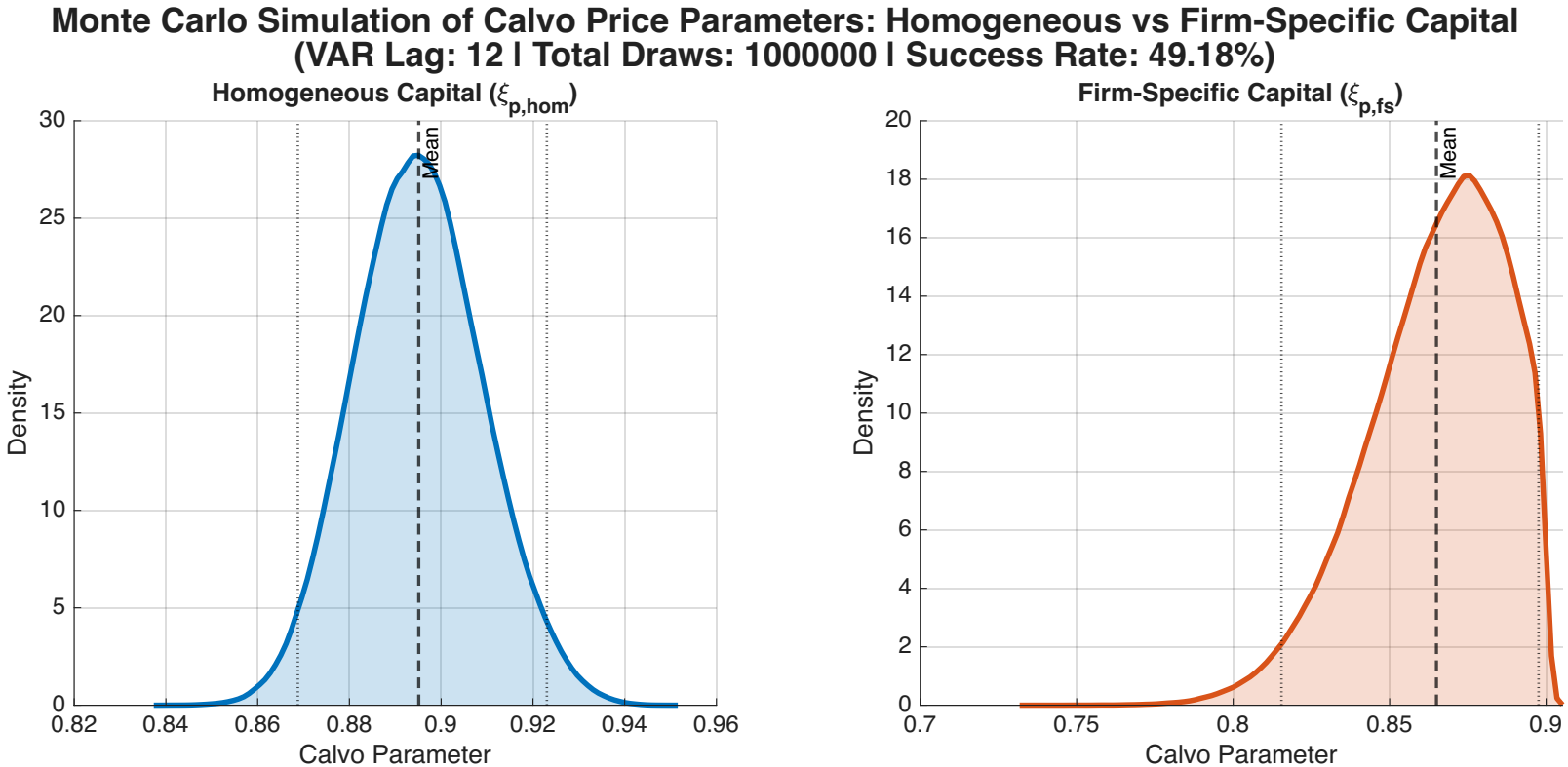


Figure 9: Calvo Price Parameter PDF from MonteCarlo Simulation; VAR(12)

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